
Celebrating 100 Years
Retirement Complex
Do Not Block
WPW
Calci-Yum
ice cream
Very Vanilla
Delicious Dark Chocolate
Non-Dairy
Let it set for 4 Weeks
MAX capacity 48
CLEAR MOVE AHEAD
I'M NEW
Be classy NOT Tachy
I've Got Rhythm
SUPPORT OUR VETS
MEDS
NEVER BEEN EASIER
I DO til I die
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2.3 - Narrow-Complex Tachycardia 3: AFib, & AFlutter Management

1) Elevated heart clock:
Supraventricular tachycardias (SVTs) typically present as narrow complex (QRS < 120ms, 3 boxes) **tachycardias (>100 bpm)** → A fib ~100-200 bpm; A flutter ~150 bpm w/ 2:1 block

2) Narrow "complex":
Supraventricular tachycardias (SVTs) typically present as **narrow complex (QRS < 120ms, 3 boxes)** tachycardias (>100 bpm) → impulse originates at/above AV node (supraventricular) to His Purkinje system of ventricles → aberrant conduction (bundle branch block, accessory pathway) may cause wide complex

3) Irregularly irregular static on screen:
AFib is caused by continuous, chaotic electrical signals from all over the atria (>350 signals/min) → some signals transmitted to ventricles through AV node (~100-200 bpm depending on refractory period of node, ~90-170 bpm in older pts) → **"irregularly irregular"** rhythm (different R-R intervals in no obvious pattern) **(AFib/ ECG)**

4) Wavy baseline static:
no distinct P waves in the setting of AFib → wavy erratic baseline **(f waves)** between the transmitted QRS complexes

5) Dog with "fluttering" ears:
no distinct P waves in the setting of AFlutter → **"sawtooth" flutter waves (F waves)**~300 bpm → typically 1:2 signals filtered through AV node (1:2 block) → regular narrow QRS complexes (ventricular rate ~150 bpm)

6) Regular intermittent jumps:
AFlutter → "sawtooth" flutter waves (F waves) ~300 bpm → typically **1:2 signals filtered through AV node (1:2 block)** → **regular narrow QRS complexes (ventricular rate ~150 bpm)**; other ratios (e.g. 1:3, 1:4) also common

7) New guy:
New onset AFib/AFlutter → hemodynamically unstable? (hypotension, AMS, HF, ischemia) → immediate synchronized cardioversion

8(P) Falling brain hat and failing heart balloon:
New onset AFib/AFlutter → **hemodynamically unstable (hypotension, AMS, HF, ischemia)** → immediate synchronized cardioversion **(AFib&AFlutter/ Plan (1))**

9(P) Synchronized sparklers:
New onset AFib/AFlutter → hemodynamically unstable (hypotension, AMS, HF, ischemia) → **immediate synchronized cardioversion (AFib&AFlutter/ Plan (2))**

10(P) Symptomatic but stable:
New onset AFib/AFlutter (hemodynamically **STABLE**) → **rate control needed (to treat tachycardia/improve hemodynamics** → ↓**symptoms**) → give AV nodal blocker (beta blocker or nondihydropyridine CCB; digoxin if ↓systolic fxn/relative hypotension) **(AFib&AFlutter/ Plan (3))**

11(P) Blocked note conduction:
New onset AFib/AFlutter (hemodynamically STABLE) → rate control needed? (to treat tachycardia/improve hemodynamics → ↓symptoms) → **give AV nodal blocker** (beta blocker or nondihydropyridine CCB; digoxin if ↓systolic fxn/relative hypotension) **(AFib&AFlutter/ Plan (4))**

12(P) Muted beta bugle + calci-yum ice cream:
New onset AFib/AFlutter (hemodynamically STABLE) → rate control needed? (to treat tachycardia/improve hemodynamics → ↓symptoms) → give AV nodal blocker (**beta blocker or nondihydropyridine CCB**; digoxin if ↓systolic fxn/relative hypotension) **(AFib&AFlutter/ Plan (5))**

13(P) Depressed loose heart accordion + foxglove flowers:
New onset AFib/AFlutter (hemodynamically STABLE) → rate control needed? (to treat tachycardia/improve hemodynamics → ↓symptoms) → give AV nodal blocker (beta blocker or nondihydropyridine CCB; **digoxin if ↓systolic fxn/relative hypotension but has ↑risk of toxicity**) **(AFib&AFlutter/ Plan (6))**

14(P) “Do no block” + wolf pennant:
Patients with **AFib + preexcitation (e.g. Wolff Parkinson White)** → wide complex, irregularly irregular rhythm → **do NOT give AV nodal blocker** (increased conduction of impulses down accessory pathway, leading to ↑ventricular rate → hemodynamic instability/VFib) → rhythm control preferred over rate control **(AFib&AFlutter/ Plan (7))**

15(P) Synchronized sparklers on dance floor:
New onset AFib/AFlutter (STABLE) → consider rate control (if sx/tachy) → **consider cardioversion (typically sync elec cardioversion** +/- antiarrhythmics) to return pt to sinus rhythm (most patients with NEW onset AFib) → anticoagulation strategy (AFib <48hrs vs >48hrs) **(AFib&AFlutter/ Plan (8))**

16(P) Synchronized sparklers with meds:
New onset AFib/AFlutter (STABLE) → consider rate control (if sx/tachy) → consider cardioversion (typically sync elec cardioversion **+/- antiarrhythmics**) to return pt to sinus rhythm (most patients with NEW onset AFib) → anticoagulation strategy (AFib <48hrs vs >48hrs) **(AFib&AFlutter/ Plan (9))**

17(P) Elderly dude with unraveling heart off of synchronized dance floor:
New onset AFib/AFlutter (STABLE) → consider rate control (if sx/tachy) → consider cardioversion (typically sync elec cardioversion +/- antiarrhythmics) to return pt to sinus rhythm (**asymptomatic elderly pts w/ comorbidities typically NOT cardioverted**→ consider long-term anticoagulation + rate control strategy) **(AFib&AFlutter/ Plan (10))**

18(P) “anti-clog” “4 weeks” “max capacity 48”:
New onset AFib/AFlutter (STABLE) → consider rate control (if sx/tachy) → consider cardioversion to return pt to sinus rhythm (most patients with NEW onset AFib) → **anticoagulation strategy** (AFib <48hrs = typically anticoagulated w/ immediate cardioversion; >48hrs = **delay cardioversion after 3-4 weeks anticoagulation** unless TEE shows no atrial clot) → anticoagulation typically contd. 4 weeks **(AFib&AFlutter/ Plan (11))**

19(P) Ultrasound speaker “move ahead”:
New onset AFib/AFlutter (STABLE) → consider rate control (if sx/tachy) → consider cardioversion to return pt to sinus rhythm (most patients with NEW onset AFib) → anticoagulation strategy (AFib <48hrs = typically anticoagulated w/ immediate cardioversion; >48hrs = delay cardioversion after 3-4 weeks anticoagulation **unless TEE shows no atrial clot, then proceed to cardioversion**) → anticoagulation typically contd. 4 weeks (**AFib&AFlutter/ Plan (12))**

20(P) Immediately on synchronized dance floor:
New onset AFib/AFlutter (STABLE) → consider rate control (if sx/tachy) → consider cardioversion to return pt to sinus rhythm (most patients with NEW onset AFib) → anticoagulation strategy (**AFib <48hrs** = typically anticoagulated w/ **immediate cardioversion**; >48hrs = delay cardioversion after 3-4 weeks anticoagulation unless TEE shows no atrial clot) → anticoagulation typically contd. 4 weeks (**AFib&AFlutter/ Plan (13))**

21(P) Anti-clog during the synchronized dance:
New onset AFib/AFlutter (STABLE) → consider rate control (if sx/tachy) → consider cardioversion to return pt to sinus rhythm (most patients with NEW onset AFib) → anticoagulation strategy (AFib <48hrs = **typically anticoagulated w/ immediate cardioversion**; >48hrs = delay cardioversion after 3-4 weeks anticoagulation unless TEE shows no atrial clot) → anticoagulation typically contd. 4 weeks **(atrial stunning + ↑risk of embolization ~1 month post procedure) (AFib&AFlutter/ Plan (14))**

22(P) Stunned on dance floor + 4 week anti-clog:
New onset AFib/AFlutter (STABLE) → consider rate control (if sx/tachy) → consider cardioversion to return pt to sinus rhythm (most patients with NEW onset AFib) → anticoagulation strategy (AFib <48hrs = typically anticoagulated w/ immediate cardioversion; >48hrs = delay cardioversion after 3-4 weeks anticoagulation unless TEE shows no atrial clot) → **anticoagulation typically contd. 4 weeks (atrial stunning + ↑risk of embolization ~1 month post procedure) (AFib&AFlutter/ Plan (15))**

23(P) Chronic grandfather clock + "recurrent" "persistent":
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider **long-term** rate vs rhythm control (↓recurrence, ↓sx, ↓risk of tachy-mediated cardiomyopathy, ↓thromboembolic events) → consider long-term anticoagulation (CHA2DS2 VASc ≥ 1) **(AFib&AFlutter/ Plan (16))**

24(P) Rate-controlling metronome vs rhythm songbook:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term **rate vs rhythm control** (↓recurrence, ↓sx, ↓risk of tachy-mediated cardiomyopathy, ↓thromboembolic events) → consider long-term anticoagulation (CHA2DS2 VASc ≥ 1)**(AFib&AFlutter/ Plan (17))**

25) "classy not tachy":
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control (↓**recurrence, ↓sx, ↓risk of tachy-mediated cardiomyopathy**, ↓thromboembolic events) → consider long-term anticoagulation (CHA2DS2 VASc ≥ 1)

26) "=" thromboembolic bird poop:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control (↓recurrence, ↓sx, ↓risk of tachy-mediated cardiomyopathy, **decreased but still present risk of thromboembolic events whether rate/rhythm control pursued**) → consider long-term anticoagulation (CHA2DS2 VASc ≥ 1)

27(P) Happy/asx elderly lady next to rate metronome:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control → **rate control (BBs, CBBs, digoxin) preferred in asx and elderly patients (↑risk of drug SE with antiarrhythmics) and those with long-standing disease (less likely to cardiovert)** → consider long-term anticoagulation (CHA2DS2 VASc ≥ 1)**(AFib&AFlutter/ Plan (18))**

28(P) Younger visitors next to rhythm songbook:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control → **rhythm control preferred** in younger patients with shorter duration of disease (more likely to cardiovert), **symptomatic patients, or those who have failed rate control** → consider long-term anticoagulation (CHA2DS2 VASc ≥ 1) (AFib&AFlutter/ Plan (19))

29(P) Symptomatic grandma and broken rate metronome next to rhythm songbook:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control → **rhythm control preferred** in younger patients with shorter duration of disease (more likely to cardiovert), **symptomatic patients, or those who have failed rate control** → consider long-term anticoagulation (CHA2DS2 VASc ≥ 1) (AFib&AFlutter/ Plan (20))

30) Purple phone, flakes next to rhythm songbook:
rhythm control agents (**class 1 = propafenone, flecainide**; class 3 = amiodarone, sotalol, dofetilide) are used to convert/maintain sinus rhythm +/- synch elec cardioversion

31) Amigo hat, “I do till I die”, soda:
rhythm control agents (class 1 = propafenone, flecainide; **class 3 = amiodarone, sotalol, dofetilide**) are used to convert/maintain sinus rhythm +/- synch elec cardioversion

32(P) Anti-clog next to rhythm songbook:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control → **if rhythm control pursued, choose anticoagulation strategy** (AFib <48hrs = typically anticoagulated w/ immediate cardioversion; >48hrs = delay cardioversion after 3-4 weeks anticoagulation unless TEE shows no atrial clot → anticoagulation typically contd. 4 weeks) → consider long-term anticoagulation (CHA2DS2 VASc ≥ 1) **(AFib&AFlutter/ Plan (21))**

33(P) Vibrating heart next to rhythm songbook:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control → **if rhythm control pursued, be aware of drug SE = PROarrhythmic effects (v tach, torsades, v fib)** → assess renal function (excretion) + avoid drugs that ↑QT + pt must immediately report sx of dizziness, syncope, dyspnea **(AFib&AFlutter/ Plan (22))**

34(P) Failing heart balloon next to rhythm songbook:

Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control → if rhythm control pursued, be aware of drug SE = **PROarrhythmic effects (v tach, torsades, v fib) → increased risk in pts with underlying heart disease**→ some drugs may be contraindicated in pts w/ hx of CAD, HF, or LVH **(AFib&AFlutter/ Plan (23))**

35(P) Ablated radio:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control → rhythm control preferred in younger patients with shorter duration of disease (more likely to cardiovert), symptomatic patients, or those who have failed rate control → **radiofrequency ablation (myocardium surrounding pulm veins) in pts who have failed/are intolerant of antiarrhythmic drugs (AFib&AFlutter/ Plan (24))**

36(P) Anti-clog next to chronic grandfather clock:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control (decreased but still present risk of thromboembolic events whether rate/rhythm control pursued) → **consider long-term anticoagulation (CHA2DS2 VASc ≥ 1) (AFib&AFlutter/ Plan (25))**

37(P) Stained ischemic hat vs leaky anti-clog:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control (decreased but still present risk of thromboembolic events whether rate/rhythm control pursued) → **consider long-term anticoagulation, balance risk of bleeding vs risk of thromboembolic event/stroke** (CHA2DS2 VASc ≥ 1) **(AFib&AFlutter/ Plan (26))**

38) Ischemic stained “CHA2DS2 VAcS” guy:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term anticoagulation based on risk of embolic stroke → CHA2DS2 VASc = Hypertension (1 point), **A**ge over 75 (2 points), **D**iabetes (1 point), previous **S**troke (2 points), **V**ascular disease (1 point), **A**ge 65-74 (1 point), and female **S**ex (1 point)

39(P) “>1” vacuum under “CHA2DS2 VAcS” guy:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term anticoagulation based on risk of embolic stroke → **CHA2DS2 VASc > high risk of embolic stroke → long-term anticoagulation (no contraindications) required to decrease stroke risk**; CHA2DS2 VASc = 1 Intermediate risk, anticoagulate case by case basis **(AFib&AFlutter/ Plan (27))**

40(P) “War” vet next to anti-clog:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control (decreased but still present risk of thromboembolic events whether rate/rhythm control pursued) → consider long-term anticoagulation, balance risk of bleeding vs risk of thromboembolic event/stroke (CHA2DS2 VASc >1 = high risk of stroke requiring anticoagulation, 1 = intermediate risk) → **warfarin** or NOACS reduce the risk of thromboembolic event ~70% **(AFib&AFlutter/ Plan (28))**

41) ==P= INteRcom II-III under “War” vet:
Warfarin is an oral anti-Vit K anticoagulant that **requires monitoring (INR 2-3, for recurrent paroxysmal/persistent AFib/AFlutter)** and may interact with other drugs/diet**(AFib&AFlutter/ Plan (29))**

42(P) No bridge:
Warfarin and NOACs typically **do NOT require bridging** when started to reduce thromboembolic risk in pts w/ recurrent paroxysmal/persistent AFib/AFlutter; warfarin req several days to take effect and may need heparin bridging in pts with especially high risk/hx of thromboembolism (e.g. known atrial clot) **(AFib&AFlutter/ Plan (30))**

43(P) Easy oral meds next to anti-clog:
Recurrent paroxysmal/persistent AFib/AFlutter (STABLE) → consider long-term rate vs rhythm control (decreased but still present risk of thromboembolic events whether rate/rhythm control pursued) → consider long-term anticoagulation, balance risk of bleeding vs risk of thromboembolic event/stroke (CHA2DS2 VASc >1 = high risk of stroke requiring anticoagulation, 1 = intermediate risk) → warfarin or **NOACS** reduce the risk of thromboembolic event ~70% **(AFib&AFlutter/ Plan (31))**

44(P) “Never been easier”:
Novel oral anticoagulants (NOACs, e.g. dabigatran, rivaroxaban, apixaban, or edoxaban) **do NOT require monitoring → convenient, effective, but expensive (AFib&AFlutter/ Plan (32))**

45(P) Fluttering dog next to rhythm songbook:
rhythm control preferred over rate control for AFlutter, specifically → radiofrequency ablation (cavotricuspid isthmus of the tricuspid annulus) tx of choice **(AFib&AFlutter/ Plan (33))**

46(P) Fluttering dog ablating radio:
rhythm control preferred over rate control for **AFlutter**, specifically → **radiofrequency ablation (cavotricuspid isthmus of the tricuspid annulus) tx of choice (AFib&AFlutter/ Plan (34))**